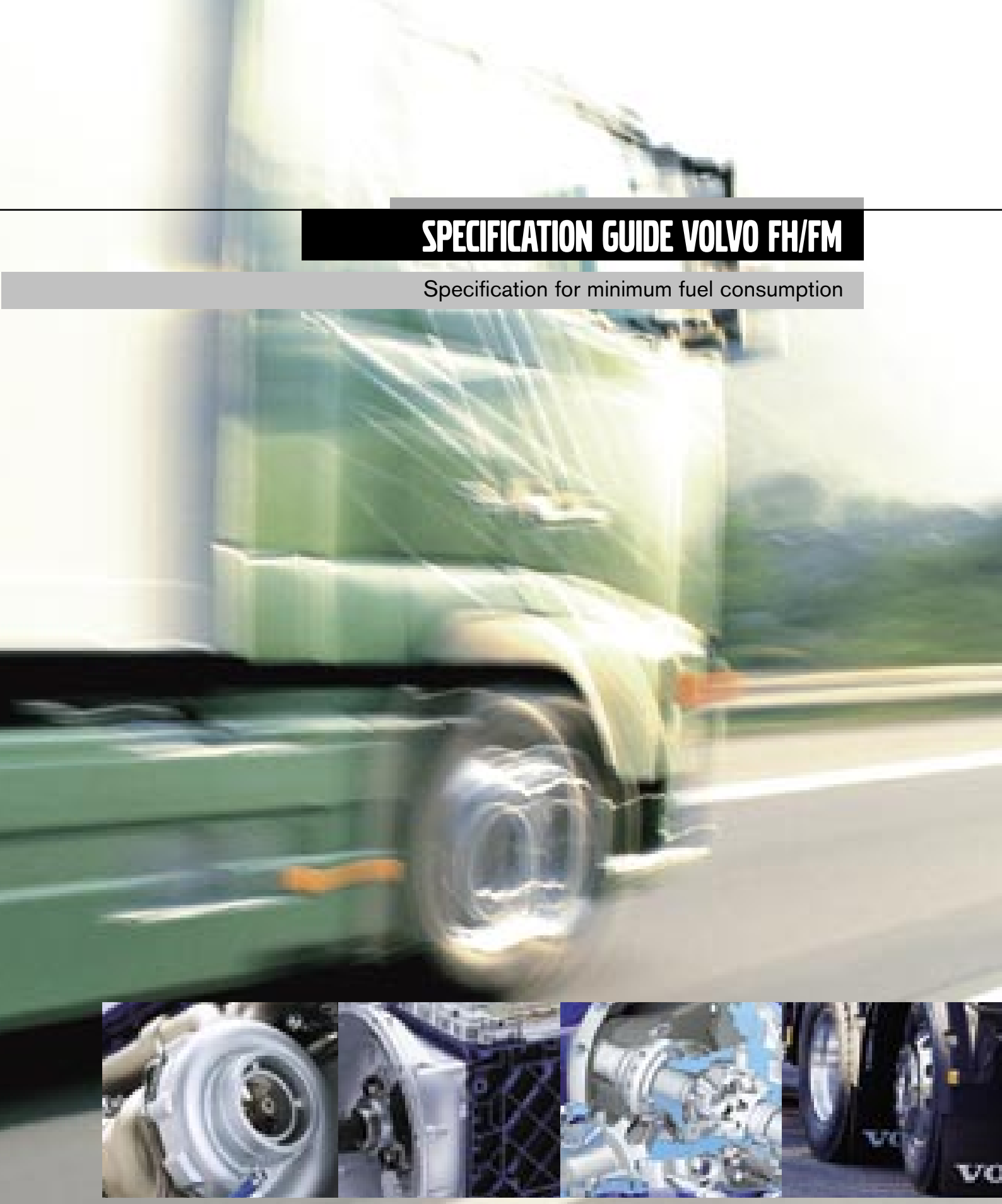
SPECIFICATION GUIDE VOLVO FH/FM

Specification for minimum fuel consumption



For internal use only

VOLVO

SPECIFICATION FOR THE OPTIMUM CONSUMPTION

Volvo FH and FM now offer more performance levels, gearbox options and driven axle ratios than ever before. With these come ever-increasing opportunities to customise the engine-gearbox-driven axle ratio-tyre configuration to suit your needs. Increased performance and a more efficient utilisation of fuel energy can also produce an increase in the average speed and lower fuel consumption – if you choose the correct specification.

The ever-increasing importance of the right specifications

Selecting the correct engine, gearbox, driven axle ratio and tyre dimensions is becoming an ever-more important factor in ensuring optimal performance. This brochure features examples of specifications that will guarantee a suitable performance level for your customers with the emphasis on fuel economy. However, this is not merely a question of specifications for performance driving and increased average speed, but of economising driving by targeting fuel consumption. The examples given are based on normal European driving conditions and certification according to Euro3.

New possibilities

The combination of D9A/D12D and I-shift offers you the opportunity to actively customise the driveline to meet the requirements of the customer. You can now choose between five Data Kits, of which three focus primarily on low fuel consumption, see pages 8 and 9. For customers involved in normal long-haul operations with a maximum gross combination weight (GCW) of 44 tonnes, you can now also offer EconomyPower – a variant of the D12D420/460, which together with the I-shift transmission reduces fuel consumption even further. You can find out more about EconomyPower on pages 10 and 11.

Low fuel consumption is not influenced solely by the properties of the driveline and the tyre dimensions, but also by low air drag. Volvo's tailor-made air deflectors are an extremely important factor in achieving fuel savings.

PERF- your most important support system
Although an ever-expanding specification base and improved and modernised product characteristics offer you more possibilities, you still need to conduct a detailed requirements analysis in order to be able to recommend the best combination. Furthermore, the list of customer requirements continues to grow, especially when it involves fuel consumption. To achieve the optimum result, it is important – vital – that you use PERF, which is Volvo's PC-based program for the analysis and selection of the correct driveline – based on the requirements of each individual transport assignment. Examples of the support offered by PERF may be found on pages 6 and 7.

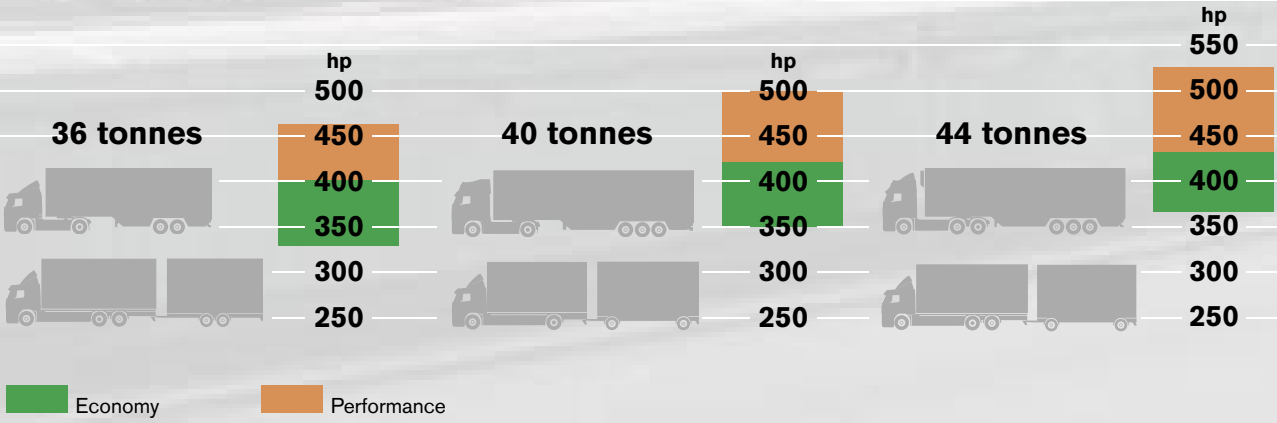
How to improve fuel economy
This brochure contains many examples of various driveline specifications that offer low fuel consumption in combination with an enjoyable driving experience. In order to achieve better fuel economy, which, at the end of the day, is the most important aspect, the customer is willing to compromise on utilising a lower average speed and employing economy driving. Note what is said about PERF and the importance of always analysing each individual case, starting with its criteria. Always remember that a high average speed and tough driving environment mean an increased performance requirement, perhaps also slower driven axle ratio, different tyre dimensions, etc., which in turn increase fuel consumption.

THE CORRECT SPECIFICATIONS

Volvo's modern engines have a high efficiency rating within the normal field of activity. As regards fuel consumption, it is essential that the engine's most fuel-efficient rpm range is used at cruising speed. Therefore, the main aim is to choose a specification that matches the economic rpm range and cruising speed (80-90 kph) of the engine. This is important since a normal long-haul transporter spends 80-90% of driving time in top gear and at cruising speed.

High torque and broad rpm range
All Volvo FH/FM engine variants are low-rpm engines, which can reach maximum torque even at a low engine rpm, while the rpm range for the maximum torque is extremely broad. Moreover, the engine responds directly to the throttle. In practice, this means better towing characteristics and quicker acceleration contribute to a more pleasurable driving experience. Quite simply, it is easier to maintain speed and tempo when driving.

Volvo engines now deliver a torque that allows the driver to utilise low and therefore more fuel-efficient engine revs at cruising speed. Electronic control of the driveline – engine + I-shift – also proves advantageous when it comes to fuel economy. The computers analyse and adjust the operating functions according to the driving conditions, which means that a driver who is in difficulty – it's not always possible to remain alert while driving – automatically receives assistance and is "corrected".



Correct engine revs at cruising speed
Today Volvo has a comprehensive range of electronically-controlled engines based on the same advanced and ground-breaking engine technology. The optimal, most fuel-efficient rpm range is somewhat lower than in previous engine generations. At a cruising speed in top gear (80-90 kph), you must ensure that the following rpm range is achieved at cruising speed for each engine variant:

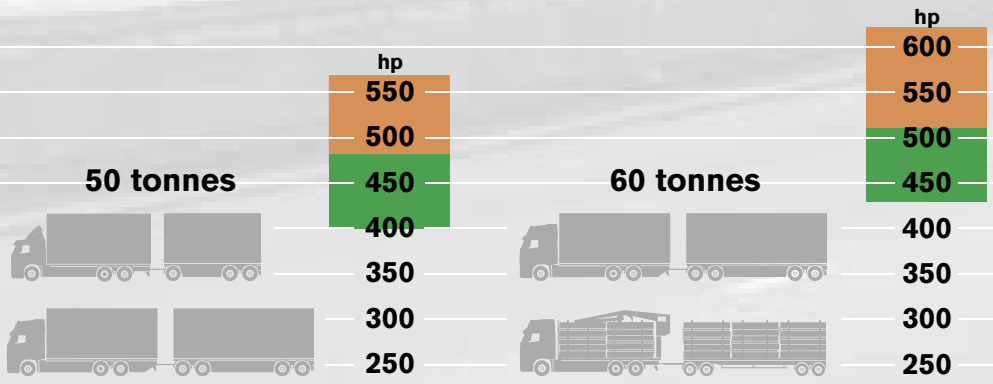
D9A-260-380 – 1450–1550 rpm
D12D-340-460 – 1350–1450 rpm
D12D-500 – 1250–1400 rpm
D16C-550, 610 – 1100–1300 rpm.

This means choosing a driven axle ratio and tyre dimensions to match these requirements. Always try to be near the lower limit of the optimal rpm range at cruising speed.

Of course, you must also take into account those requirements and recommendations regarding hill-climbing ability, driving time in top gear, the number of gear changes per 100 km, etc. This is primarily a question of engine performance, which is determined by the gross combination weight of the vehicle combination, see below.

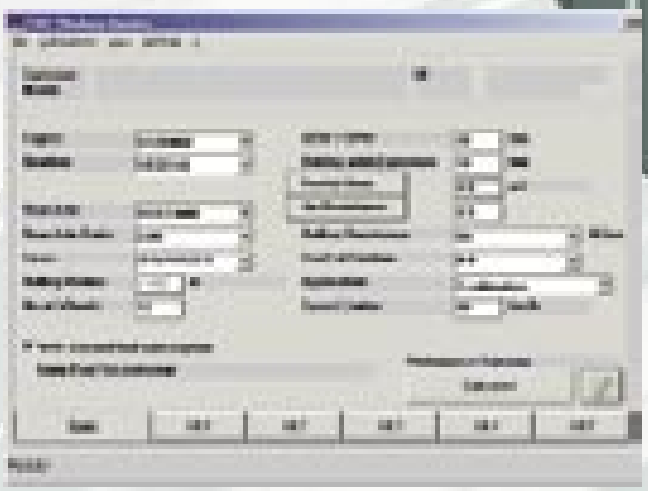
Increased power requirement
There are several reasons why power outputs are increasing. Two main reasons are tougher competition in the haulage industry and the demands of the customer for faster transporters. Naturally, another reason is because the maximum speed of the trucks is limited nowadays, and the only way to increase the average speed is to increase speed on uphill sections. Volvo's auxiliary brake system and electronic control also afford opportunities to increase speeds safely. This performance requirement is primarily connected to the gross combination weight. The current rule of thumb states that a truck suitable for economy driving requires 8-10 hp/tonne, while a performance truck requires 10-12 hp/tonne. This is shown in outline in the tables below.

Hill-climbing ability is the answer
Once you have chosen the engine power output, axle ratio and tyre dimensions, etc. in PERF, you get an account overview of the driveline capacity. Hill-climbing ability in top gear is an indicator of both the vehicle's driveability and fuel consumption. "fuel economy" vehicle combinations used under normal conditions should be able to manage a 1.2% gradient in top gear. The equivalent value for the "performance truck" is 1.5%. The driving time in top gear can be used as an indicator of driveability. In this case, at least 80% of the driving time should be spent in top gear.



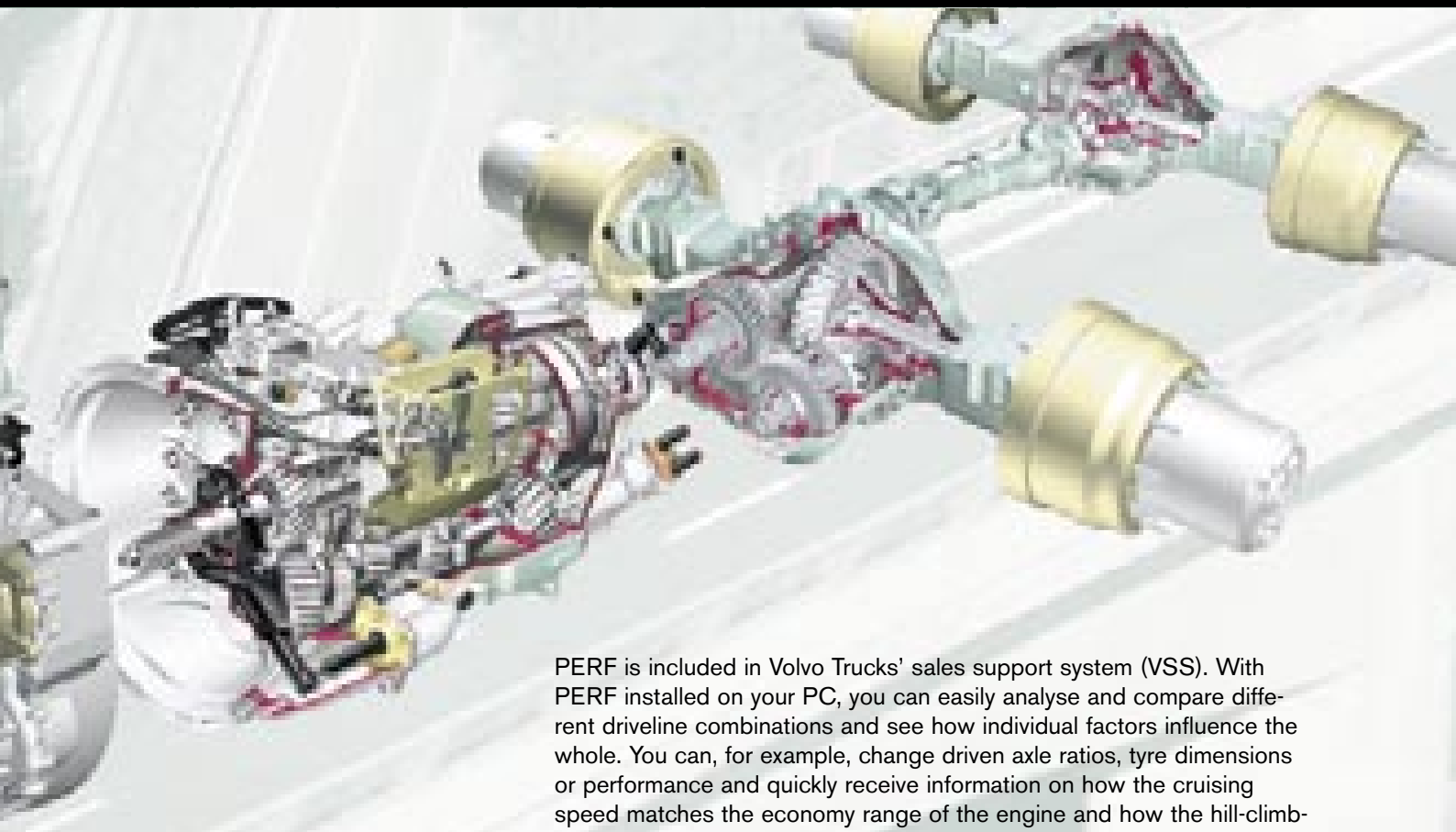
PERF – PERFORMANCE CALCULATIONS

PERF is quite simply the first four letters of the word PERFormance. PERF is a PC-based analytical tool that offers you the opportunity to evaluate in detail the capacity of the driveline and the criteria for low fuel consumption – based on the requirements and criteria of each individual transport assignment.



Main window
Example of main window with selected specifications.

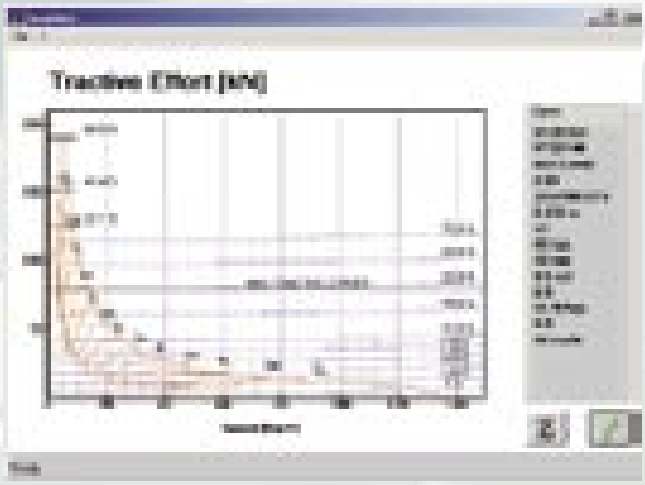
Performance summary
Result window with performance summary displaying, for example, vehicle speed vs economy rpm, maximum vehicle speed, maximum hill-climbing capacity in various conditions and starting capacity.



PERF is included in Volvo Trucks' sales support system (VSS). With PERF installed on your PC, you can easily analyse and compare different driveline combinations and see how individual factors influence the whole. You can, for example, change driven axle ratios, tyre dimensions or performance and quickly receive information on how the cruising speed matches the economy range of the engine and how the hill-climbing capacity is altered.

The program is based on window menus from which you simply choose between different options by scrolling through them and filling in the various fields with facts about the relevant transport operation. The results are displayed using graphs and in tabular form, both on screen and as a printout (see the example in this document).

Product comparison
Result window showing various specifications and the effect of individual factors on the result.



Pulling power
Graphic window with diagram of pulling power – maximum pulling power in kN for various gears, the speed range of various gears, hill-climbing capacity in % for each gear.

SMART CONTROL AND STEERING

In combination with the I-shift transmission, Volvo's D9A and D12D engines create the most modern and harmonised driving power system in the world. The advantages to the customer are low fuel consumption and low operating costs. Furthermore, the driver always works in a relaxed working environment with a high level of safety. The various engine and gearbox functions are determined by the vehicle's operating conditions, where the data control units quickly adapt performance and function to suit each new operating situation.

Tailor-made control with the right Data Kit

There are five different Data Kits to choose between for the I–shift transmission. This gives you the opportunity to recommend to the customer the option that will best meet their needs. The various characteristics of each Data Kit are presented in the table and are briefly summarised below.

The Data Kits are:

Basic (1) – is the standard version providing basic functions.

Fuel economy (2) – is good for long-haul operations, where long distances take their toll on fuel economy.

Comfort (3) – makes trucks more nimble and easier to handle when driving in built-up areas.

Fuel economy and Comfort (4) – for those customers who want to enjoy the advantages of both Fuel economy and Comfort.

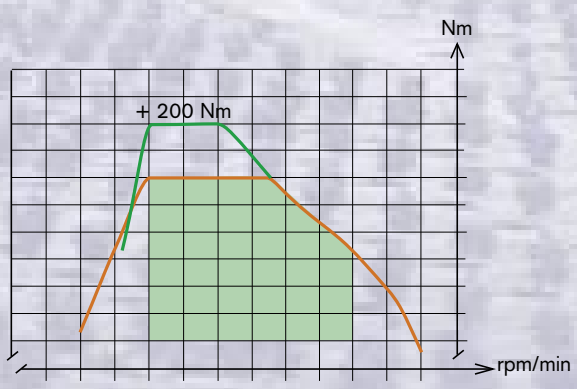
EconomyPower (5) – for those customers who want to make the most of their fuel and have D12D420/460. Read more about this overleaf.

Engine	D9A D12D				D12D 420/460
Data Kit	**1**	**2**	**3**	**4**	**5**
Power Mode Std	●				
Automatic Mode Std	●				
Power Mode Prof		●	●	●	●
Automatic Mode Prof		●	●	●	●
EcoRoll		●		●	●
Smart Cruise Control		●		●	●
Performance Shift		●	●	●	●
Aut. EBR control		●	●	●	●
Brake Mode		●	●	●	●
Smooth range shift			●	●	●
Clutch Disc Services			●	●	●
Oil Temp			●	●	●
Kick down			●	●	●
High Torque Direct Gear					●

Power Mode Prof	- automatic return to the E-program when extra power is not needed.
Automatic Mode Prof	- automatic adjustment of starting gear.
EcoRoll	- automatic and controlled switching to neutral once the accelerator pedal is released.
Smart Cruise Control	- auxiliary brakes are not activated automatically in certain situations.
Performance Shift	- quicker and gentler gearchanging function.
Aut. EBR control	- intelligent activation of engine brake.
Brake Mode	- optimises VEB/retarder function, selects gear for high engine revs.
Smooth range shift	- quieter and gentler range changing.
Clutch Disc Services	- indicates that the clutch disc needs replacing.
Oil Temp	- displays the gearbox oil temperature in the driver information display unit.
Kick down	- maximum acceleration when the accelerator pedal is pressed to the floor.
High Torque Direct Gear	- automatic higher torque in top gear, D12D420/460.

THE ULTIMATE COMBINATION – D12D420/460+I-SHIFT

Where the main requirement for quick long-haul operations is low fuel consumption, you can offer the customer the ultimate solution – the EconomyPower Data Kit. This is available in both the FH12 and FM12 for up to 44 tonnes GCW, with engine variants D12D420/460 and in combination with the I-shift transmission. EconomyPower delivers increased performance in top gear, + 200 Nm, which quite simply makes it possible to drive for longer periods in top gear and therefore save fuel.



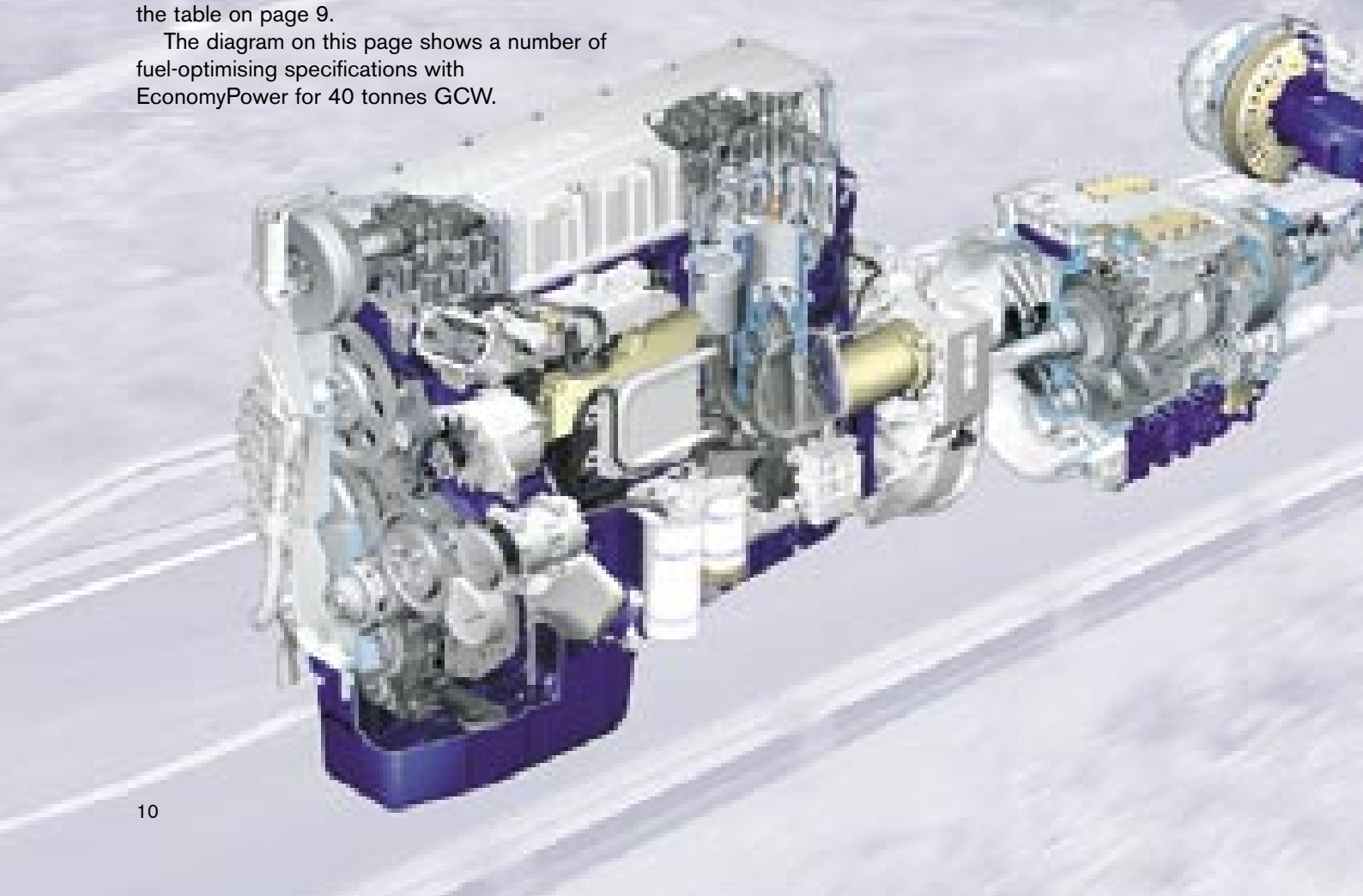
EconomyPower – increased torque in top gear (skeleton diagram).

More power requires less fuel

The solution is as smart as it is simple. Once the truck drives in top gear, the electronic system provides the driver with a further 200 Nm in the rpm range normally used at cruising speed. Once the I-shift switches to top gear, the engine's data control unit simply adjusts the fuel injection in order to increase the power. The extra 200 Nm in top gear results in a better hill-climbing capacity, which in turn means longer periods in top gear – the power passes straight through the gearbox – this produces minimal losses. Each gear change missed also means a fuel saving.

The other functions included in EconomyPower are shown in the table on page 9.

The diagram on this page shows a number of fuel-optimising specifications with EconomyPower for 40 tonnes GCW.



EconomyPower – current variants

D12D420/460	I-shift	RS1356SV	2.50 :1 2.79 :1 3.10 :1	295/60R22.5 315/70R22.5 315/80R22.5
		RSS1344B	2.64 :1 2.85 :1 3.08 :1	295/60R22.5 315/70R22.5 315/80R22.5

D12D-420, RSS1344B

Recommended rpm range in top gear at cruising speed. Engine economy range.

			Hill-climbing capacity in top gear Max %	at 80 kph %
2,64:1	295/60R22.5	1100 1200 1300 1400 1500 1600 1700 rpm/min; 70 80 90 100 110 kph	2.2	2.1
2,85:1	295/80R22.5	1100 1200 1300 1400 1500 1600 1700; 70 80 90 100 110	2.0	2.0
2,85:1	315/70R22.5	1100 1200 1300 1400 1500 1600 1700; 70 80 90 100 110	2.1	2.1
2,85:1	315/80R22.5	1100 1200 1300 1400 1500 1600 1700; 70 80 90 100 110	1.9	1.9

D12D-460, RSS1356SV

Recommended rpm range in top gear at cruising speed. Engine economy range.

			Hill-climbing capacity in top gear Max %	at 80 kph %
2,50:1	295/60R22.5	1100 1200 1300 1400 1500 1600 1700 rpm/min; 70 80 90 100 110 kph	2.3	2.2
2,79:1	295/80R22.5	1100 1200 1300 1400 1500 1600 1700; 70 80 90 100 110	2.2	2.2
2,79:1	315/70R22.5	1100 1200 1300 1400 1500 1600 1700; 70 80 90 100 110	2.3	2.3
2,79:1	315/80R22.5	1100 1200 1300 1400 1500 1600 1700; 70 80 90 100 110	2.1	2.1

SINGLE VEHICLE

4×2 19 t

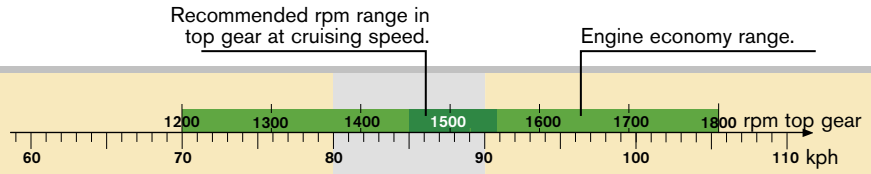
FM9-260	VT1708B/I-shift	RSS1344B	3,36:1	315/80R22.5
FM9-300	VT1708B/I-shift	RSS1344B	3,36:1	315/80R22.5
FM9-340	VT2009B/I-shift	RSS1344B	3,08:1	315/70R22.5
FM9-380	VT2009B/I-shift	RSS1344B	3,08:1	315/70R22.5

6×2 26 t

FM9-260	VT1708B/I-shift	RSS1344B	3,36:1	315/80R22.5
FM9-300	VT1708B/I-shift	RSS1344B	3,36:1	315/80R22.5
FM9-340	VT2009B/I-shift	RSS1344B	3,08:1	315/70R22.5
FM9-380	VT2009B/I-shift	RSS1344B	3,08:1	315/70R22.5
FM9-340	VT2009B	RS1352HV	3,61:1	13R22.5
FM9-380	VT2009B	RS1352HV	3,61:1	13R22.5
FM12-340	VT2009B/I-shift	RSS1344B	3,08:1	315/80R22.5
FM12-380	VT2009B/I-shift	RSS1344B	3,08:1	315/80R22.5
FM12-380	VT2009B	RS1370HV	3,61:1	13R22.5
FM12-420	VT2009B	RS1370HV	3,61:1	13R22.5

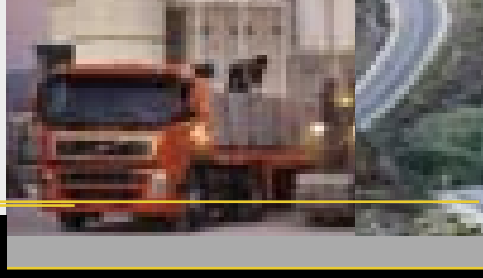
6×4, 8×4 26/32 t

FM9-340	VT1708B	RTS2370A	3,4:1	315/80R22.5
FM9-380	VT2009B	RTS2370A	3,4:1	315/80R22.5
FM9-380	VT2009B	RT2610HV	3,61:1	13R22.5
FM12-340	VT2214B	RTS2370A	3,4:1	315/80R22.5
FM12-380	VT2214B	RTS2370A	3,4:1	315/80R22.5
FM12-420	VT2214B	RTS2370A	3,4:1	315/80R22.5
FM12-420	VTO2214B	RT2610HV	4,12:1	13R22.5



Hill-climbing capacity in top gear Max %	at 80 kph %		
2,7	2,5		
3,2	3,0		
3,7	3,5		
4,0	3,8		
1,9	1,7		
2,2	2,0		
2,5	2,4		
2,8	2,6		
2,6	2,4		
2,8	2,6		
2,6	2,4		
2,9	2,7		
3,2	3,0		
3,6	3.3		
6×4 26 t	8×4 32 t	6×4 26 t	8×4 32 t
2,6	2,0	2,4	1,9
2,8	2,2	2,7	2,1
2,7	2,1	2,5	1,9
2,9	2,2	2,7	2,0
3,2	2,5	3,0	2,3
3,6	2,8	3,3	2,6
2,9	2,2	2,7	2,1

The hill-climbing ability in top gear for a single-body vehicle should be 2%.

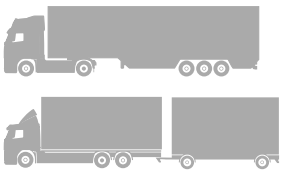


VEHICLE COMBINATIONS 34-36, 40 TONNES

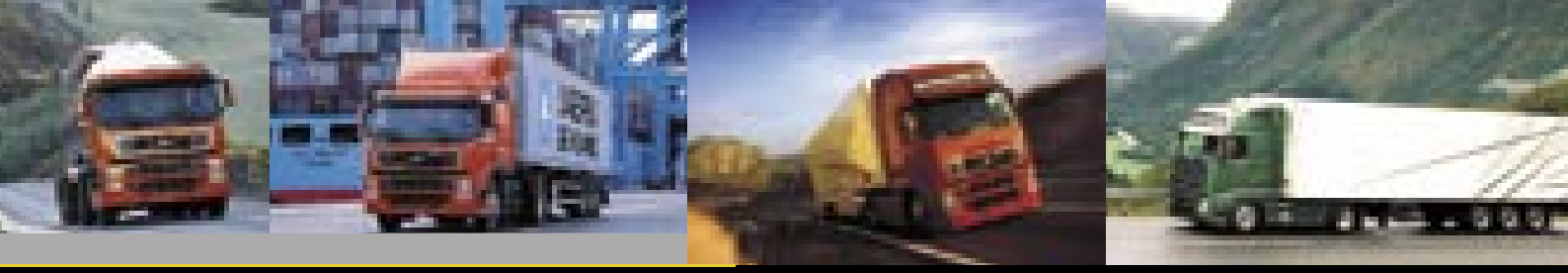
4×2 34-36 t

FM9-340	VT2009B/I-shift	RSS1344B	3,36:1	315/80R22.5
FM9-380	VT2214B/I-shift	RSS1344B	3,36:1	315/80R22.5
FM9-380	VT2214B/I-shift	RSS1344B	3,08:1	315/70R22.5
FM12-340/380	VT2009B/I-shift	RSS1344B	3,08:1	315/80R22.5

6×2 40 t

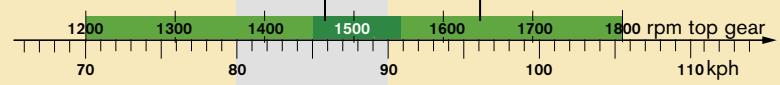


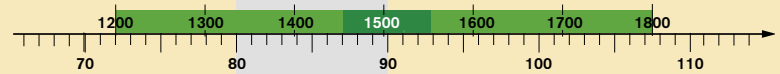
FM9-380	VT2214B/I-shift	RSS1344B	3,08:1	315/70R22.5
FH/FM12-380	VT2214B/I-shift	RSS1344B	3,08:1	315/70R22.5
FH/FM12-380	VT2214B/I-shift	RSS1344B	3,08:1	315/80R22.5
FH/FM12-420	VT2214B/I-shift	RSS1344B	3,08:1	315/80R22.5
FH/FM12-460	VT2214B/I-shift	RSS1344B	3,08:1	315/80R22.5
FH/FM12-420	VT2214B/I-shift	RSS1344B	2,85:1	315/70R22.5
FH/FM12-460	VT2214B/I-shift	RSS1344B	2,85:1	315/70R22.5
FH/FM12-420	VT2214B/I-shift	RSS1344B	2,64:1	295/60R22.5
FH12-500	VT2514B/I-shift	RS1356SV	2,79:1	315/70R22.5
FH/FM12-460	VT2514B/I-shift	RS1356SV	2,50:1	295/60R22.5
FH16-550	VT2814B	RS1356SV	2,79:1	315/80R22.5
FH16-610	VT2814B	RS1356SV	2,79:1	315/80R22.5

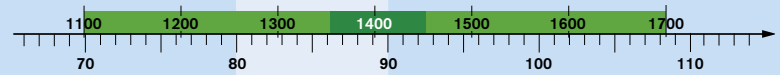


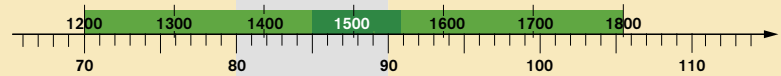
Recommended rpm range in top gear at cruising speed.

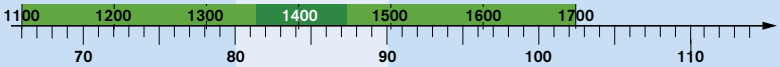
Engine economy range.

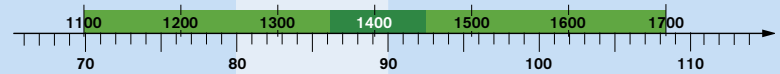


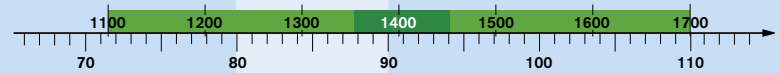


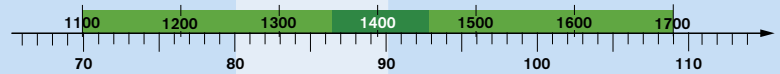


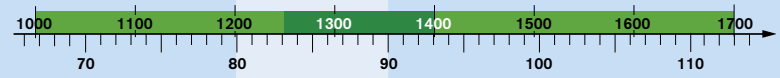


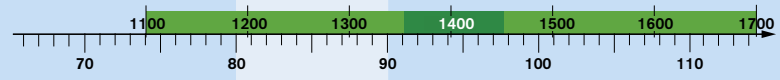










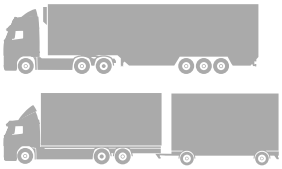


Hill-climbing capacity in top gear

Max %	at 80 kph %
1,8	1,7
1,9	1,8
1,8	1,8
1,7	1,6
1,9	1,8
1,6	1,5
1,9	1,7
1,7	1,6
1,9	1,8
2,2	2,1
1,8	1,8
2,1	2,0
1,9	1,8
2,4	2,2
2,0	1,9
2,3	2,2
2,7	2,6

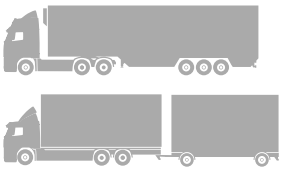
VEHICLE COMBINATION 44 TONNES

6×2 44 t

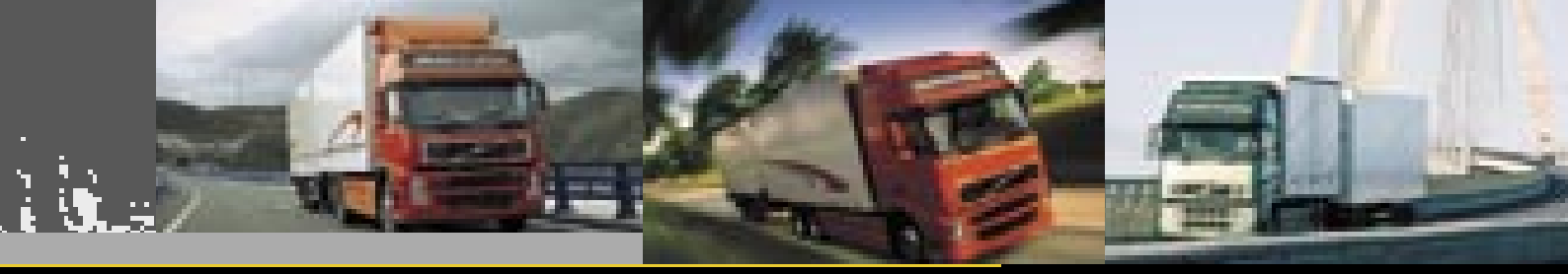


FH/FM12-420	VT2214B/I-shift	RSS1344B	3,08:1	315/80R22.5
FH/FM12-460	VT2214B/I-shift	RSS1344B	3,08:1	315/80R22.5
FH/FM12-420	VT2214B/I-shift	RSS1344B	2,85:1	315/70R22.5
FH/FM12-460	VT2214B/I-shift	RSS1344B	2,85:1	315/70R22.5
FH12-500	VT2514B/I-shift	RS1356SV	2,79:1	315/70R22.5

6×4 44 t

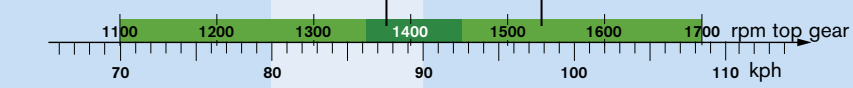


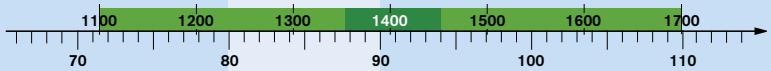
FH/FM12-420	VT2214B/I-shift	RTS2370A	3,09:1	315/80R22.5
FH/FM12-460	VT2214B/I-shift	RTS2370A	3,09:1	315/80R22.5
FH12-500	VT2514B/I-shift	RTS2370A	3,09:1	315/80R22.5
FH16-550	VT2814B	RS1356SV	2,79:1	315/80R22.5
FH16-610	VT2814B	RS1356SV	2,79:1	315/80R22.5
FH16-550	VT2814B	RTS2370A	2,83:1	315/80R22.5
FH16-610	VT2814B	RTS2370A	2,83:1	315/80R22.5

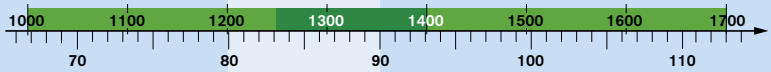


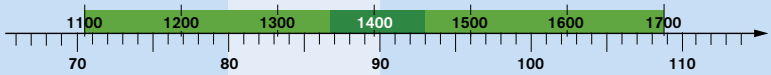
Recommended rpm range in top gear at cruising speed.

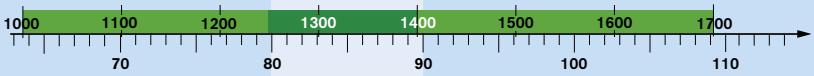
Engine economy range.

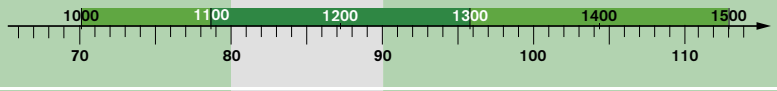


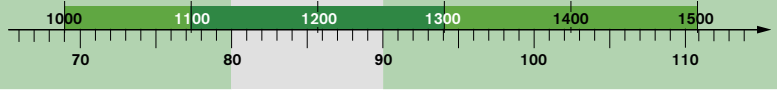












Hill-climbing capacity in top gear

Max %	at 80 kph %
1,7	1,6
1,9	1,8
1,6	1,5
1,9	1,8
2,1	2,0
1,6	1,5
1,8	1,7
2,1	2,0
2,1	2,0
2,4	2,4
2,1	2,0
2,4	2,3

VEHICLE COMBINATION 60 TONNES

6×2 60 t

FH12-460	VT2514B/GT	RS1356SV	3,10:1	315/70R22.5
FH12-500	VT2514B/GT	RS1356SV	3,10:1	315/80R22.5
FH16-550	VTO2814B	RS1370HV	3,76:1	315/80R22.5
FH16-610	VTO2814B	RS1370HV	3,76:1	315/80R22.5

6×4 60 t

FH/FM12-460	VT2514B/GT	RTS2370A	3,40:1	315/80R22.5
FH12-500	VT2514B/GT	RTS2370A	3,09:1	315/80R22.5
FH16-550	VT2814B	RTS2370A	3,09:1	315/80R22.5
FH16-610	VTO2814B	RTS2370A	3,78:1	315/80R22.5

6×4 60 t

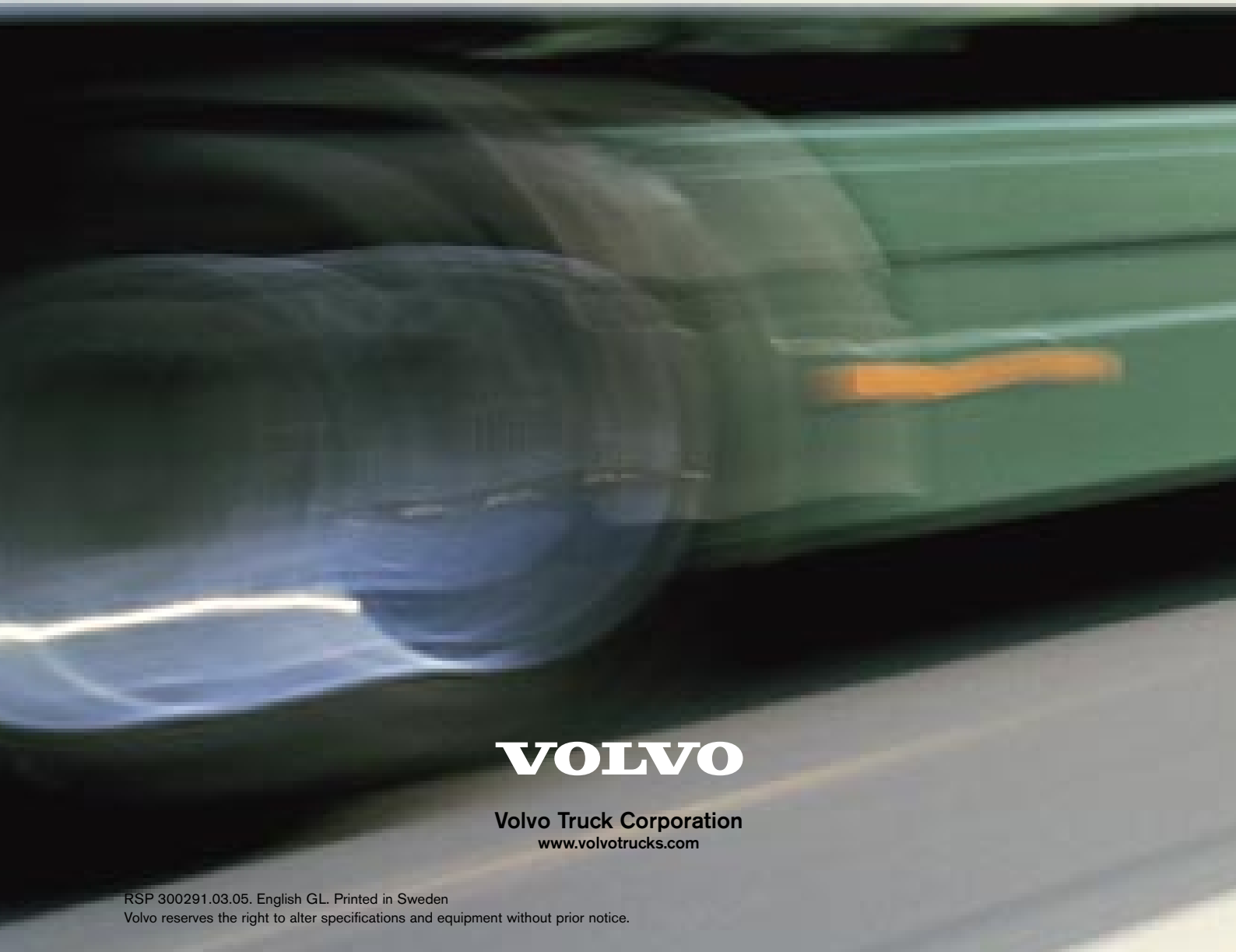
Timber

FH12-500	VTO2814B	RT2610HV	4,12:1	315/80R22.5
FH16-550	VTO2814B	RTS2370A	3,78:1	315/80R22.5
FH16-610	VTO2814B	RTS2370A	3,78:1	315/80R22.5
FH16-550	VTO2814B	RT2610HV	3,76:1	315/80R22.5
FH16-610	VTO2814B	RT2610HV	3,76:1	315/80R22.5

Recommended rpm range in top gear at cruising speed.

Engine economy range.

	Hill-climbing capacity in top gear Max %	at 80 kph %
1100 1200 1300 1400 1500 1600 1700 rpm top gear; 60 70 80 90 100 110 kph	1.4	1.2
1000 1100 1200 1300 1400 1500 1600 1700; 60 70 80 90 100 110	1.5	1.3
1000 1100 1200 1300 1400 1500; 60 70 80 90 100 110	1.3 1.6	1.2 1.4
1100 1200 1300 1400 1500 1600 1700; 60 70 80 90 100 110	1.3	1.2
1000 1100 1200 1300 1400 1500 1600 1700; 60 70 80 90 100 110	1.3	1.1
1000 1100 1200 1300 1400 1500; 60 70 80 90 100 110	1.4	1.3
1000 1100 1200 1300 1400 1500; 60 70 80 90 100 110	1.6	1.4
1000 1100 1200 1300 1400 1500 1600 1700; 60 70 80 90 100 110	1.4	1.2
1000 1100 1200 1300 1400 1500; 60 70 80 90 100 110	1.4 1.7	1.3 1.6
1000 1100 1200 1300 1400 1500; 60 70 80 90 100 110	1.4 1.5	1.3 1.4



VOLVO

Volvo Truck Corporation
www.volvotrucks.com

RSP 300291.03.05. English GL. Printed in Sweden
Volvo reserves the right to alter specifications and equipment without prior notice.